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RESEARCH ARTICLE

Myocardial infarction detection using ITD, DWT and deterministic learning based on ECG signals

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Abstract

Nowadays, cardiovascular diseases (CVD) is one of the prime causes of human mortality, which has received tremendous and elaborative research interests regarding the prevention issue. Myocardial ischemia is a kind of CVD which will lead to myocardial infarction (MI). The diagnostic criterion of MI is supplemented with clinical judgement and several electrocardiographic (ECG) or vectorcardiographic (VCG) programs. However the visual inspection of ECG or VCG signals by cardiologists is tedious, laborious and subjective. To overcome such disadvantages, numerous MI detection techniques including signal processing and artificial intelligence tools have been developed. In this study, we propose a novel technique for automatic detection of MI based on disparity of cardiac system dynamics and synthesis of the standard 12-lead and Frank XYZ leads. First, 12-lead ECG signals are synthesized with Frank XYZ leads to build a hybrid 4-dimensional cardiac vector, which is decomposed into a series of proper rotation components (PRCs) by using the intrinsic time-scale decomposition (ITD) method. The novel cardiac vector may fully reflect the pathological alterations provoked by MI and may be correlated to the disparity of cardiac system dynamics between healthy and MI subjects. ITD is employed to measure the variability of cardiac vector and the first PRCs are extracted as predominant PRCs which contain most of the cardiac vector's energy. Second, four levels discrete wavelet transform with third-order Daubechies (db3) wavelet function is employed to decompose the predominant PRCs into different frequency bands, which combines with three-dimensional phase space reconstruction to derive features. The properties associated with the cardiac system dynamics are preserved. Since the frequency components above 40 Hz are lack of use in ECG analysis, in order to reduce the feature dimension, the advisable sub-band (D4) is selected for feature acquisition. Third, neural networks are then used to model, identify and classify cardiac system dynamics between normal (healthy) and MI cardiac vector signals. The difference of cardiac system dynamics between healthy control and MI cardiac vector is computed and used for the detection of MI based on a bank of estimators. Finally, experiments are carried out on the PhysioNet PTB database to assess the effectiveness of the proposed method, in which conventional 12-lead and Frank XYZ leads ECG signal fragments from 148 patients with MI and 52 healthy controls were extracted. By using the tenfold cross-validation style, the achieved average classification accuracy is reported to be 98.20%. Results verify the effectiveness of the proposed method which can serve as a potential candidate for the automatic detection of MI in the clinical application.

Keywords Myocardial infarction (MI) detection · Electrocardiographic (ECG) · Intrinsic time-scale decomposition (ITD) · Discrete wavelet transform (DWT) · Cardiac system dynamics · Neural networks · Deterministic learning

Introduction

Cardiovascular diseases (CVD) are the most common cause of human mortality (Maheshwari et al. 2015). Measurement of timing intervals and amplitudes of the electrocardiographic (ECG) or vectorcardiographic (VCG) waves provides the morphological characteristics. It can serve as the fundamental feature in CVD diagnostics.

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Myocardial ischemia results in a characteristic pattern of metabolic and ultrastructural changes that will lead to myocardial infarction (MI). Pathological alterations provoked by MI cause slow conduction by increasing axial resistance on coupling between cells. This issue may cause abnormal patterns in the dynamics of the tip of the cardiac vector (Hejc et al. 2015). Hence, analysis of the morphology of ECG or VCG recordings plays an important role in monitoring the variation of dynamics in the pathological alterations of cardiac patterns caused by MI. In order to diagnose CVD, the standard 12-lead ECG has been regarded as the gold standard for monitoring and analyzing alternations in the cardiac activity (Wehr et al. 2006). From the perspective of medical science, cardiologists have been accustomed to the standard 12-lead system due to its noninvasiveness, inexpensiveness and availability for CVD diagnosis. Traditionally, detection of MI based on the visual inspection of ECG or VCG signals by cardiologists is tedious, laborious and subjective (Lee et al. 2017). In addition, it requires expertise in the analysis of long-term ECG or VCG signals (Sahoo et al. 2017). In those application scenarios absence of experts, for example, in emergency situations, computer-aided automatic detection of MI becomes significant.

To overcome the aforementioned disadvantages of the standard 12-lead ECG system, strategies can be taken in two aspects: (1) reducing the number of leads; and (2) employing artificial intelligence tools to analyze and classify cardiac patterns. In the first aspect, several studies have been proposed to reduce the number of leads that can reconstruct the 12-lead ECG signals without loss of significant information in diagnosis. These methods can be categorized into two types: using subsets of the 12-lead and using special leads (Tomasic and Trobec 2013; Lee et al. 2017). For the first category, subsets of the leads of the 12-lead ECG are used to reconstruct the standard 12-lead ECG (Scherer et al. 1989; Nelwan et al. 2000, 2004; Drew et al. 2002; Atoui et al. 2010; Nallikuzhy and Dandapat 2017). The missing precordial leads are reconstructed from the remaining leads by using either a personalized or a universal transformation. Smith et al. (2021) proposed a new method to reconstruct a full 12-lead ECG from a reduced lead set. Five-electrode placement was used to generate leads I, II, III, aVL, aVR, aVF and V2. These seven leads served as inputs to a time-delay neural network which derived the remaining five precordial leads (V1, V3–V6). For the second category, special leads are used based on the orthogonal lead system. For example, the orthogonal Frank XYZ leads and EASI leads are well-known for use of fewer leads compared to the widely used 12-lead ECG systems, since it is convenient to implement and can capture more information than Holter leads (Tomasic and Trobec 2013; Dawson et al. 2009; Maheshwari et al. 2016; Frank 1954; Dower 1968).

In the second aspect to overcome the disadvantages, ECG or VCG signal based arrhythmia or MI detection techniques including signal processing and artificial intelligence tools have been developed, such as linear (Kim et al. 2007; Lee et al. 2007) and nonlinear methods (Acharya et al. 2017d; Babaoglu et al. 2010), Wavelet Transform (WT) (Kumar et al. 2016a), Complex Wavelet Transform (CWT) (Banerjee and Mitra 2013; Yochum et al. 2016), Pitch Synchronous Wavelet Transform (PSWT) (Jagannath and Selvakumar 2014), Discrete Wavelet Transform (DWT) (Khambhati and Patel 2017), Kalman filtering (KF) (Chen et al. 2017), Least Mean Squares algorithm (LMS) (Roonizi and Sassi 2015), ensemble learning (Kumar et al. 2016b; Kumar and Inbarani 2017), Artificial Neural Networks (ANN) (Shadmand and Mashoufi 2016), Adaptive Neuro-fuzzy Inference System (ANFIS) (Homaeinezhad et al. 2012), support vector machine (SVM) (Babaoglu et al. 2010), and deep learning (Mathews et al. 2018; Acharya et al. 2017a, b, c; Tan et al. 2018; Yildirim 2018; Baloglu et al. 2019; Jahmunah et al. 2019; Sharma and Acharya 2019; Plawiak and Acharya 2019; Bhurane et al. 2019). For example, Varatharajan et al. (2018) used linear discriminant analysis (LDA) to reduce the features presented in the ECG signal, which followed with a SVM model with a weighted kernel function for the classification of cardiac arrhythmia. Plawiak (2018) used the spectral power density to enhance feature performance extracted from ECG signals and employed genetic algorithms for feature selection. Then several machine learning tools including SVM, k-nearest neighbor, ANN, have been used for the classification of cardiac disorders. Kumar and Inbarani (2017) used DWT to extract effective feautures. Then neighborhood rough set was employed for the ECG signal classification. Acharya et al. (2017c) used ECG signals of two seconds' and five seconds' durations as the input of convolutional neural network (CNN) with eleven-layer to detect different types of cardiac arrhythmia. There is no need for the pre-processing of QRS detection. Kumar et al. (2017) used flexible analytic wavelet transform (FAWT) to decompose ECG beats into subband signals. Then sample entropy was computed as features and fed into random forest (RF), J48 decision tree, back propagation neural network (BPNN), and least-squares support vector machine (LS-SVM) classifiers for detection of MI. Yildirim (2018) decomposed ECG signals into frequency sub-bands at different scales, which were used as the input of long-short term memory networks (LSTMs) for ECG signal classification. Most of the classification methods use explicit time-plane features such as ST segment, R height, T height, etc. Apart from the measurement accuracy issues of the extracted features, a large feature set is obtained when time-plane features are used for classification. As a result, rule mining techniques

are employed for feature set reduction, which increases the computational complexity. In addition, for MI detection, some available methods in the literature need to delineate various time-domain fiducial markers of ECG, like Q-wave, R-wave, T-wave, and ST-segment. The proposed technique is independent of these time-domain fiducial markers of ECG, which makes performance of the proposed method independent of the accuracy of QRS-complex, T-wave, or ST-segment detection. This permits the proposed technique to work effectively both in elevated and non-elevated ST-segment MI. Tripathy et al. (2019) proposed an approach for the detection of MI pathology using the multiresolution analysis of 12-lead ECG signals. It was based on the use of Fourier-Bessel series expansion-based empirical wavelet transform (FBSE-EWT) for the time-scale decomposition of 12-lead ECG signals. For each lead ECG signal, nine subband signals were evaluated using FBSE-EWT. The statistical features such as the kurtosis, the skewness, and the entropy were evaluated from the subband signals of each ECG lead. Deep layer least-square support-vector machine (DL-LSSVM) was used for the detection of MI from the feature vector of 12-lead ECG. Zhang et al. (2021) utilized tensorization based on DWT to construct an effective ECG tensor containing multi-dimensional association information from 12-lead ECG signals. Parallel Factor Analysis was developed to automatically extract the low-dimensional and highly recognizable lead characteristic features of the tensor. A bagged decision tree was then constructed for MI detection. Hussein et al. (2021) proposed a scheme to observe the changes in ischemic-related ECG components (ST segment and PR segment) by way of the Choi-Williams time-frequency distribution to extract ST and PR features. These extracted features were mapped to a multiclass SVM classifier for MI detection. Jahmunah et al. (2022) extracted the ECG beats using an R peak detection algorithm. The beats were then fed to DenseNet and CNN models separately for MI detection. An enhanced class activation mapping (CAM) technique called Grad-CAM was subsequently applied to the outputs of both models to enable visualization of the specific ECG leads and portions of ECG waves that were most influential for the predictive decisions made by the models.

While the above-mentioned techniques in the literature with time or frequency features have achieved great improvements in MI detection, there still remain several unsolved issues in MI detection. Because of the redundant and discrete-time characteristics of ECG and VCG signals, traditional methods such as the Fourier transform (FT) (i.e., spectral domain), are unable to analyze the nonstationary ECG and VCG signal. They fail to provide any information about the time of occurrence of frequency components. Numerous methods with combination of time and frequency domains and nonlinear analysis have been developed to handle the problem (Raj and Ray 2017). Specifically, for the nonlinear analysis, parameters extracted through different types of entropies (Acharya et al. 2018), Lyapunov exponent (Ubeyli 2009), local fractal dimension (Mishra and Raghav 2010), higher order spectra (HOS) cumulants (Martis et al. 2014), recurrence quantification analysis (RQA) (Yang 2010) and Hurst exponent (Bhaduri and Ghosh 2016), have been employed for automatic detection of abnormal ECG or VCG signals. Despite the fact that the above-mentioned methods have achieved respectable classification accuracy, the potential of nonlinear methods has not been thoroughly investigated. This includes the nonlinear features and the modeling and identification algorithms for nonlinear cardiac system dynamics. Considering the characteristics that the ECG signal is highly random, nonlinear, nonstationary and non-Gaussian in nature (Sivakumar and Nedumaran 2018), self-adaptive signal processing methods, such as empirical mode decomposition (EMD) (Huang et al. 1998; Huang and Kunoth 2013), local mean decomposition (LMD) (Park et al. 2011) and intrinsic time-scale decomposition (ITD) (Frei and Osorio 2007), can be employed to extract effective and predominant features from ECG or VCG signals (Ghobadi Azbari et al. 2016; Lu et al. 2016). In comparison to EMD and LMD, ITD significantly improves the computational efficiency. In addition, with high decomposition efficiency and frequency resolution, ITD can help decompose a complex signal into several proper rotation components (PRCs) and a baseline signal. This leads to the accurate extraction of the dynamic features of nonlinear signals.

Although so much work exists on MI detection, it is still necessary to further explore in this field. For one reason, the performance needs to be improved for modern diagnosis of myocardial infarction. For another reason, analysis of ST-segment elevation, as well as the high apex and inversion of T waves, and the appearance of pathological Q waves, is substantially more popular than the analysis of long-duration ECG signal fragments, which is error-prone and may not achieve satisfactory diagnostic performance due to high beat-to-beat variability of these features among individuals. These motivate us to carry out research on a new solution of MI detection using new features extracted from long-duration continuous ECG signals. Inspired by the core idea of signal decomposition and feature extraction (Altan et al. 2019), in the present work, we develop a novel algorithm to compute the representative features based on ITD and DWT algorithms and the synthesis of 12-lead and Frank XYZ leads. We hypothesize that these features reflect the abnormal alterations in the dynamics of the cardiac vector and can achieve high sensitivity and specificity simultaneously as a discriminator of MI, which might be more accurate than conventional algorithms.

The key contributions of this study lie in six aspects: (1) we use the synthesis of 12-lead and Frank XYZ leads to build a new 4-dimensional cardiac vector which may fully reflect the pathological alterations provoked by MI. (2) ITD method is employed to measure the variability of cardiac vector and the first proper rotation components (PRCs) are extracted as predominant PRCs which contain most of the cardiac vector's energy. (3) Discrete wavelet transform (DWT) decomposes the predominant PRCs into different frequency bands, which are used to construct the reference variables. (4) 3D phase space of the predominant PRCs components is reconstructed, in which the properties associated with the cardiac system dynamics are preserved. (5) Cardiac system dynamics can be modeled and identified using neural networks, which employ the ED of 3D PSR of the reference variables as input features. (6) The difference of cardiac system dynamics between healthy control and MI cardiac vector is computed and used for the detection of MI based on a bank of estimators. In the present study, feature extraction, training and classification constitute the main framework. Extracted features through ITD, DWT and PSR are fed into the neural networks for the modelling and identification of cardiac system dynamics. If the extracted features are not representative enough, it will influence the following training and classification performance. Since the difference of dynamics between normal (healthy) and MI cardiac vectors is applied for the classification task, the modelling precision will influence the classification performance.

Method

In this section, we propose a method to discriminate between normal and MI cardiac vectors using the information obtained from nonlinear cardiac system dynamics for the detection of MI. The flowchart of the proposed algorithm is illustrated in Fig. 1.

ECG database

In this study we utilize the popular and publicly available PTB database which has been acquired at the Department of Cardiology of University Clinic Benjamin Franklin in Berlin, Germany, and has been provided to the users of PhysioNet (Bousseljot et al. 1995; Goldberger et al. 2000). The database contains 549 records from 290 subjects, including 148 patients with MI and 52 healthy control (HC) subjects. Each subject is represented by one to five records. Hence there are 367 and 78 heart beat records for MI and HC subjects, respectively. Each record includes 15 simultaneously measured signals: the conventional 12 leads (i, ii, iii, avr, avl, avf, v1, v2, v3, v4, v5, v6) together with the 3 Frank XYZ leads ECGs (vX, vY, vZ). Each signal was digitized at 1000 samples per second.

The characteristics of the used ECG data obtained from PTB database is presented in Table 1.

Synthesis of 12-lead and Frank XYZ leads

First, transform the 12-lead ECG into 3-dimensional VCG since VCG signals and 12-lead ECG signals can be linearly transformed into each other without loss of useful information content pertaining to the cardiac system dynamics (Dower and Machado 1979). The transformation relationship is shown as follows (Kors et al. 1990):

$$\begin{cases} V_1 = 0.38x_1 - 0.07x_2 - 0.13x_7 + 0.05x_8 - 0.01x_9 + 0.14x_{10} + 0.06x_{11} + 0.54x_{12} \\ V_2 = -0.07x_1 + 0.93x_2 + 0.06x_7 - 0.02x_8 - 0.05x_9 + 0.06x_{10} - 0.17x_{11} + 0.13x_{12} \\ V_3 = 0.11x_1 - 0.23x_2 - 0.43x_7 - 0.06x_8 - 0.14x_9 - 0.20x_{10} - 0.11x_{11} + 0.31x_{12} \end{cases} \quad (1)$$

where x_1 to x_{12} represent the 12 leads (i, ii, iii, avr, avl, avf, v1, v2, v3, v4, v5, v6), respectively, V_1, V_2, V_3 represent the transformed 3 VCG signals, respectively. For the Frank XYZ leads,

$$V_4 = \sqrt{X^2 + Y^2 + Z^2} \quad (2)$$

where X, Y and Z represent the lead value of three corresponding axes, V_4 represents the VCG voltage magnitude of the 3 Frank leads by taking the square root of the sum of the squares for the three individual axes.

Here we build a new 4-dimensional cardiac vector: $V = [V_1, V_2, V_3, V_4]^T$, which contains useful information content from the 12-lead and Frank XYZ leads. It reduces the dimension of feature vectors instead of using 15 leads ECG signals directly. In addition, the novel cardiac vector may fully reflect the pathological alterations provoked by MI and may be correlated to the disparity of cardiac system dynamics between HC and MI subjects. The examples of the 4-dimensional cardiac vectors from HC and MI subjects are demonstrated in Fig. 2.

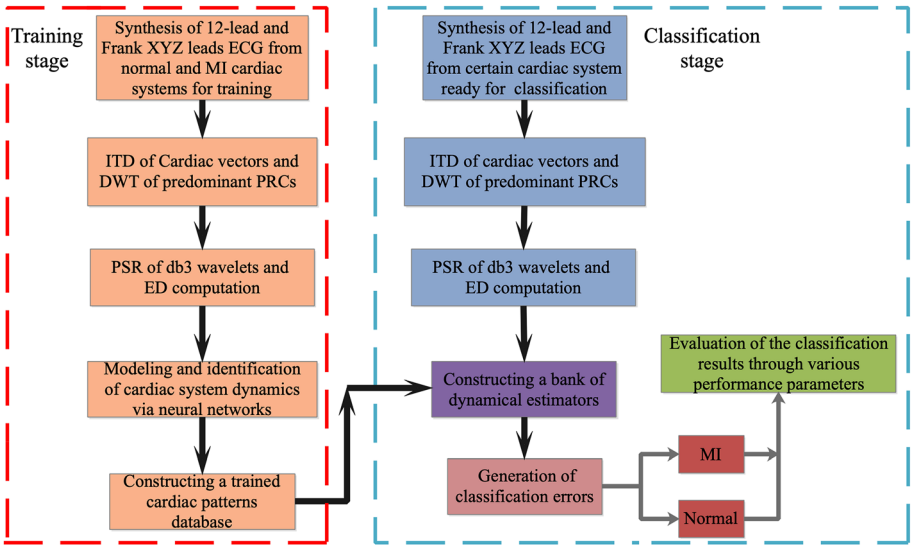


Fig. 1 Flowchart of the proposed method for the MI classification using syntheses of lead system, ITD, DWT, PSR, ED and neural networks

Table 1 Description of the PTB database with the selected ECG signals

Characteristics	HC	MI
Age range (years)	17–81	36–86
Average age (years)	43.4	60.4
Number of participants	52	148
Number of heart beat records	78	367
Ratio of male to female	3	2.89

Intrinsic time-scale decomposition (ITD)

Intrinsic time-scale decomposition (ITD) is suitable for analyzing nonstationary and nonlinear signals such as the ECG and VCG signals. Without resorting to the spline interpolation to signal extrema and sifting in mono-component separation, it decomposes a signal into proper rotation components (PRCs) that are suitable to calculate the instantaneous frequency and amplitude, based on the baseline defined via linear transform. The obtained decomposition result precisely preserves the temporal information of each component regarding signal critical points and riding waves, with a time resolution equal to the time scale of the occurrence of extrema in the raw signal (Feng et al. 2016). Based on the single wave analysis, it extracts accurately the inherent instantaneous amplitude and frequency/phase information and other relevant morphological features (Frei and Osorio 2007).

For a time series signal $I(t)$, define the operator L to extract the baseline signal from $I(t)$ and the residual signal is called the proper rotation component (PRC). The decomposed signal $I(t)$ can be expressed as

$$I(t) = LI(t) + (1 - L)I(t) = B(t) + H(t) \tag{3}$$

where $B(t)$ is the baseline signal and $H(t)$ is the proper rotation.

The decomposition procedure of a nonlinear signal can be summarized by the following steps:

- *Step 1* Find the local extrema of the signal $I(t)$, denoted by I_k, and the corresponding occurrence time instant $\tau_k, k = 0, 1, 2, \cdots$. For convenience $\tau_0 = 0$.

- *Step 2* Suppose the operators $B(t)$ and $H(t)$ are given over the interval $[0, \tau_k]$, and $I(t)$ is set on the interval $t \in [0, \tau_{k+2}]$. Then on the interval $[\tau_k, \tau_{k+1}]$ between two adjacent extrema I_k and I_{k+1}, the piecewise baseline extraction operator is defined as

$$LI(t) = B(t) = B_k + \left(\frac{B_{k+1} - B_k}{I_{k+1} - I_k}\right) \times (I(t) - I_k), t \in [\tau_k, \tau_{k+1}], \tag{4}$$

where

$$B_{k+1} = \beta\left[I_k + \left(\frac{\tau_{k+1} - \tau_k}{\tau_{k+2} - \tau_k}\right)(I_{k+2} - I_k)\right] + (1 - \beta)I_{k+1}, \tag{5}$$

and $0 < \beta < 1$, typically $\beta = 0.5$.

- *Step 3* After extracting the baseline signal, the operator Θ for extracting the residual signal as PRCs is defined as

$$\Theta I(t) \equiv (1 - L)I(t) = I(t) - B(t) \tag{6}$$

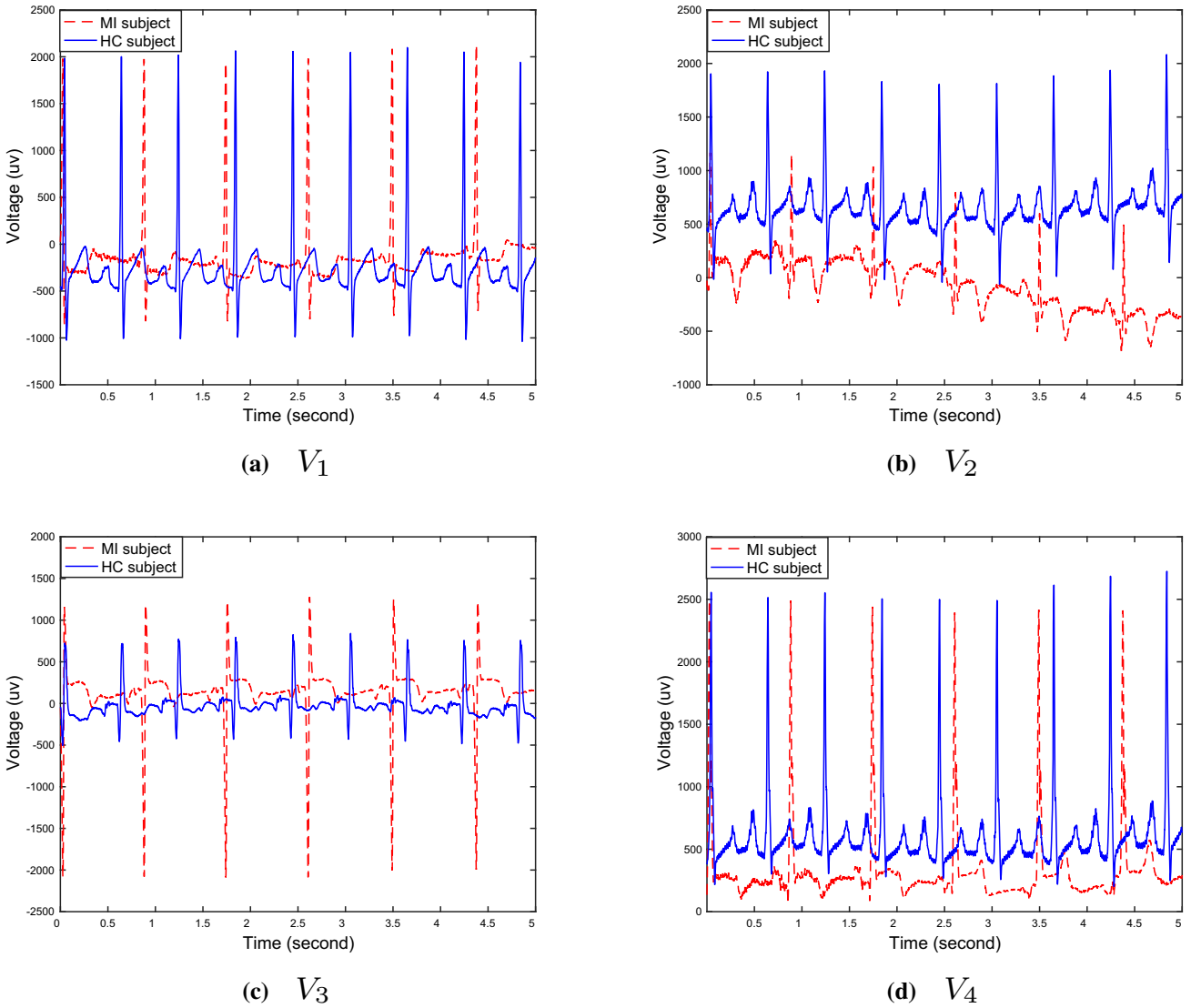


Fig. 2 Examples of the synthesis of 4-dimensional cardiac vector: $V = [V_1, V_2, V_3, V_4]^T$ in morphological variations between HC and MI subjects

According to the definition, the PRC is a riding wave with the highest frequency on the baseline. Therefore, ITD separates the PRC in a frequency order from high to low. In addition, the PRC is obtained directly by subtracting the baseline from the input signal, without resorting to any sifting within each iterative decomposition. Thus, ITD has low computational complexity, and more importantly, avoids the smoothing of transients and time-scale smearing due to repetitive sifting (Feng et al. 2016).

Take the baseline $B(t)$ as the input signal $I(t)$, and repeat steps (1)–(3), until the baseline becomes a monotonic function or a constant. Eventually, the raw signal will be decomposed into PRCs and a trend (Feng et al. 2016)

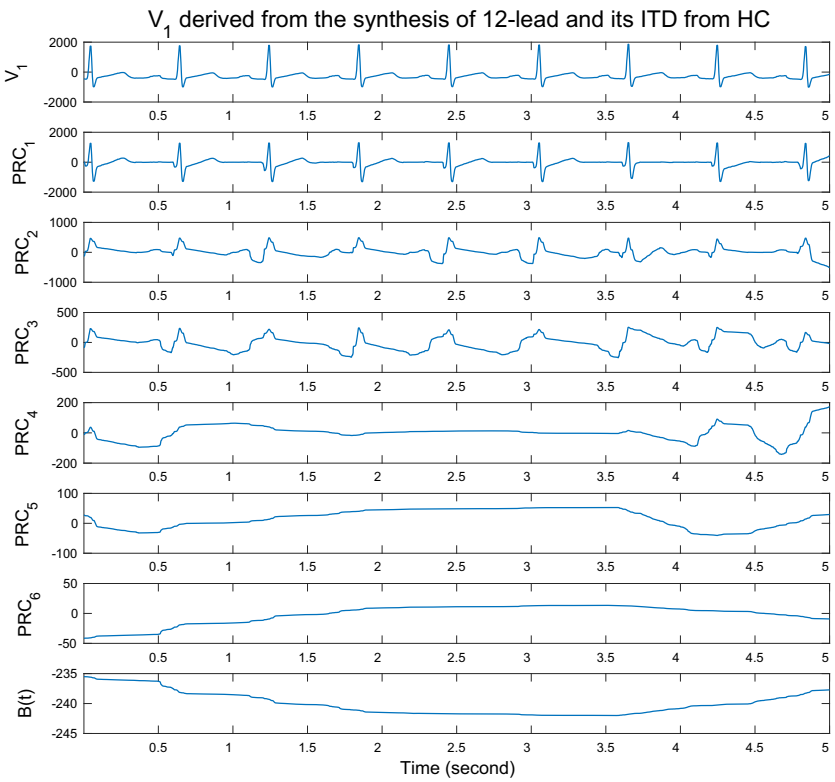
$$I(t) = \sum_{i=1}^{\rho} H^i(t) + B^{\rho}(t), \tag{7}$$

where ρ is the decomposition level.

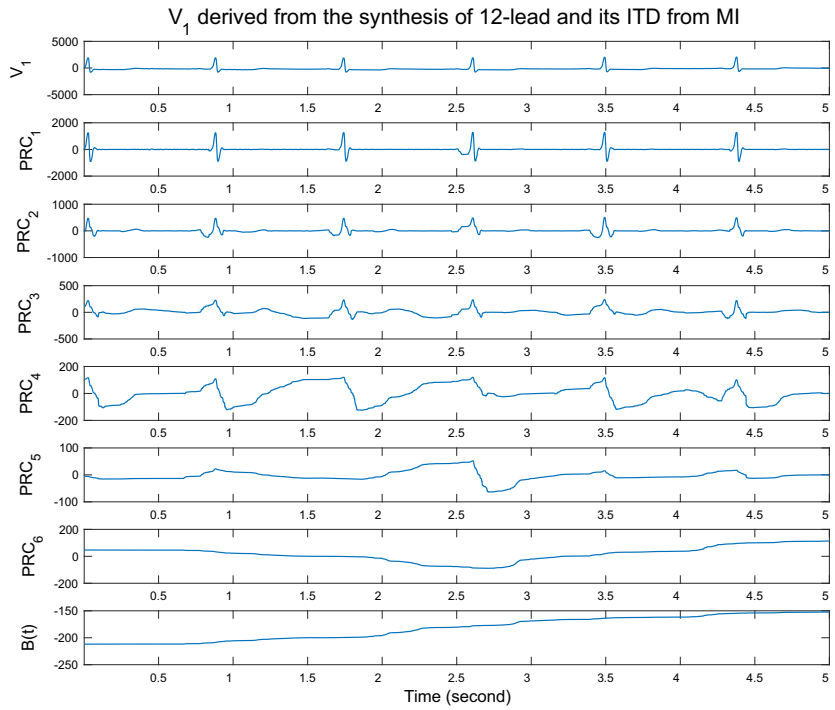
Samples of the ITD of the subvector V_1 from the HC and MI subjects are demonstrated in Fig. 3. The remainders of the cardiac vectors V_2, V_3 and V_4 and their ITD from the HC and MI subjects are demonstrated in "Fig. 7 of the Appendix".

Discrete wavelet transform (DWT)

Wavelet transform is an effective time-frequency tool for the analysis of non-stationary signals. Discrete Wavelet Transform (DWT) is a procedure for the decomposition of input signal $H(t)$ ($H(t)$ is the PRC of the cardiac vector in this work) into sets of function, called wavelets, by scaling and shifting of mother wavelet function. Consequently, the decomposition i.e. set of wavelet coefficients are formed.



(a) Original V_1 vector and its ITD from HC.



(b) Original V_1 vector and its ITD from MI.

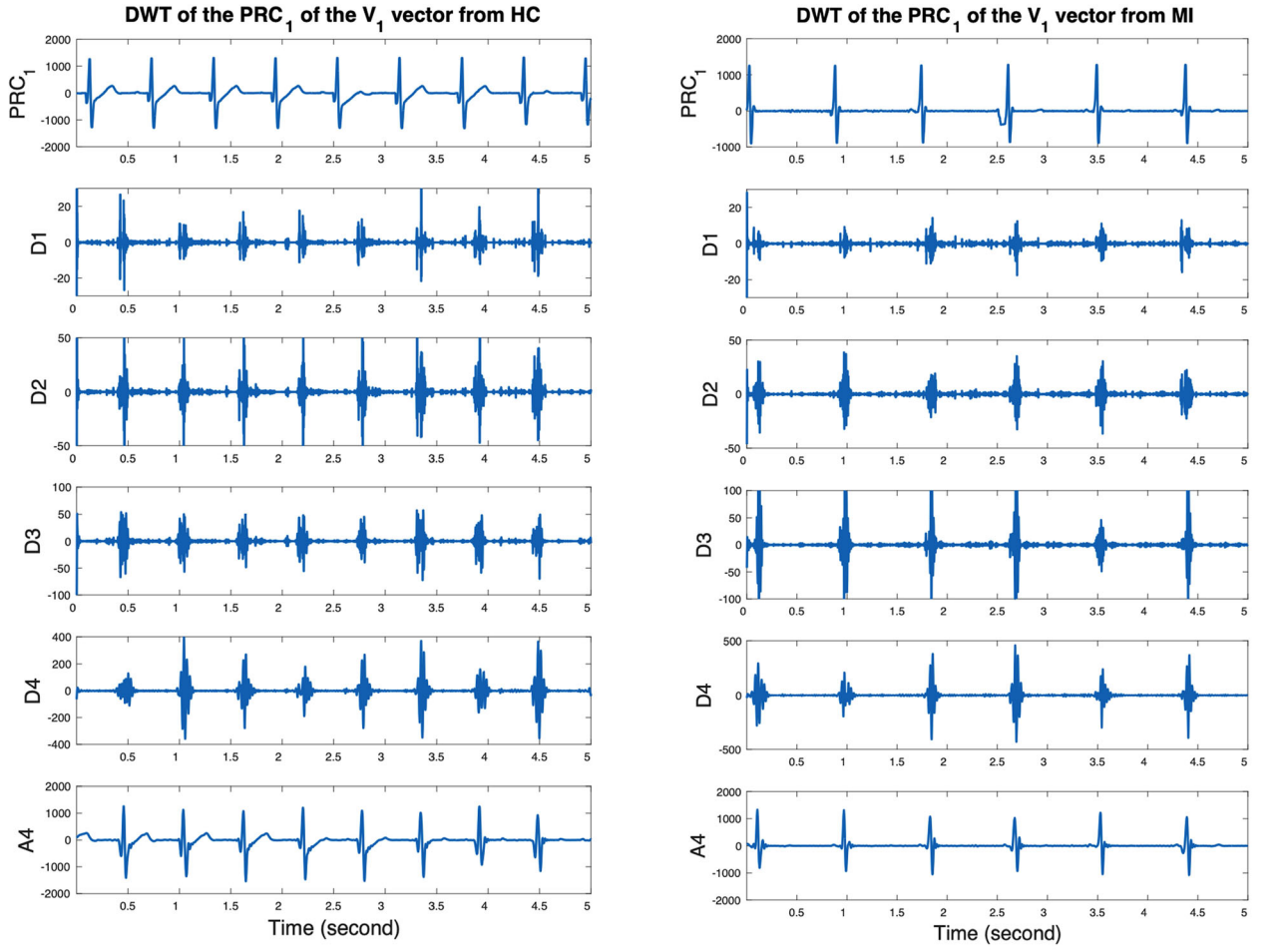
Fig. 3 Samples of the ITD of the subvector V_1 from the HC and MI subjects

To accomplish this, the signal $H(t)$ can be reconstructed as linear combination of wavelets and weighting wavelet coefficients. The setting of appropriate wavelet function and the number of decomposition levels is of great importance for correctly reconstructing the signal $H(t)$. It was reported that most of the energy of the ECG signal lies between 0.5 Hz and 40 Hz (Thakor et al. 1984), for which the wavelet coefficients is concentrated in the lower sub-bands. In order to extract five physiological cardiac vector bands, four levels DWT with third-order Daubechies (db3) wavelet function have been used (Table 2 represents the frequency distribution of the DWT-based coefficients of the PRCs of the cardiac vector at 1000 Hz), from which the choice of the mother wavelet is supported by many works in literature (Vavadi et al. 2010; Tawfik et al. 2016; Li et al. 2017). Figure 4 shows samples of V_1 vector of HC and MI subjects and their decomposed frequency bands of predominant PRCs. The remainders of the four levels DWT of PRC_1 of V_2, V_3 and V_4 from the HC and MI subjects are demonstrated in "Fig. 8 of the Appendix".

Since the frequency components above 40 Hz are lack of use in ECG analysis, in order to reduce the feature dimension, the advisable sub-band (D4) is selected for feature acquisition. The detailed information of levels l, 2 and 3 (sub-bands D1, D2 and D3) are discarded since the frequencies covered by these levels are higher than frequency content of the cardiac vector.

Table 2 Frequency band of PRCs of the cardiac vector using fourth level decomposition of DWT

Decomposition levels	Sub-bands	Frequency range (Hz)
1	D1	250–500
2	D2	125–250
3	D3	62.5–125
4	D4	31.25–62.5
4	A4	0–11.25



(a) DWT of PRC_1 of V_1 from HC subject. **(b)** DWT of PRC_1 of V_1 from MI subject.

Fig. 4 Samples of four levels DWT of PRC_1 of the subvector V_1 from HC and MI subjects

Phase space reconstruction (PSR)

It is sometimes necessary to search for patterns in a time series and in a higher dimensional transformation of the time series (Sun et al. 2015). Phase space reconstruction is a method used to reconstruct the so-called phase space. The concept of phase space is a useful tool for characterizing any low-dimensional or high-dimensional dynamic system. A dynamic system can be described using a phase space diagram, which essentially provides a coordinate system where the coordinates are all the variables comprising mathematical formulation of the system. A point in the phase space represents the state of the system at any given time (Sivakumar 2002; Lee et al. 2014). Every db3 wavelet function of the PRC of the cardiac vector can be written as the time series vector $V = \{v_1, v_2, v_3, \ldots, v_K\}$, where K is the total number of data points. The phase space can be reconstructed according to (Lee et al. 2014):

$$Y_j = (V_j, V_{j+\tau}, V_{j+2\tau}, \ldots, V_{j+(d-1)\tau}) \tag{8}$$

where $j = 1, 2, \ldots, K-(d-1)\tau$, d is the embedding dimension of the phase space and τ is a time lag. It is worthwhile to mention that the properties associated with the cardiac dynamics are preserved in the reconstructed phase space.

The behaviour of the signal over time can be visualized using PSR (especially when $d = 2$ or 3). In this work, we have confined our discussion to the value of embedding dimension $d = 3$, because of their visualization simplicity. In addition, different studies have found this value to best represent the attractor for human biological system (Venkataraman and Turaga 2016; Som et al. 2016). For τ, we either use the first-zero crossing of the autocorrelation function for each time series or the average τ value obtained from all the time series in the training dataset using the method proposed in (Michael 2005). In this study, we consider the values of time lag $\tau = 1$ to test the classification performance. PSR for $d = 3$ has been referred to as 3D PSR.

Reconstructed phase spaces have been proven to be topologically equivalent to the original system and therefore are capable of recovering the nonlinear dynamics of the generating system (Takens 1980; Xu et al. 2013). This implies that the full dynamics of the cardiac system are accessible in this space. For this reason, features extracted from it can potentially contain more and/or different information than the common features extraction method (Chen et al. 2014).

3D PSR is the plot of three delayed vectors V_j, V_{j+1} and V_{j+2} to visualize the dynamics of human cardiac system. Euclidian distance (ED) of a point (V_j, V_{j+1}, V_{j+2}), which is the distance of the point from origin in 3D PSR and can be defined as (Lee et al. 2014)

$$ED_j = \sqrt{V_j^2 + V_{j+1}^2 + V_{j+2}^2} \tag{9}$$

Feature extraction and selection

In order to obtain more efficient features, this paper proposes the following extraction scheme.

(1) ITD of the cardiac vector and derivation of predominant PRCs. The signals obtained by ITD method, which are a series of decomposing signals, cannot be directly used to classify because of the high feature dimension. To solve this problem, the Pearson's correlation coefficient is calculated to measure the correlation between the first six PRCs and the original cardiac vector. The PRC component with higher correlation coefficient is more highly correlated to the original signal, which means the signal energy is mostly concentrated in this PRC as well. In the present study most of the energy is concentrated in PRC_1 component, which have the most important information from the cardiac vector and are considered to be the predominant PRCs (seen from Table 3).

(2) Four levels DWT is employed to decompose the predominant PRCs into different frequency bands, in which third-order Daubechies (db3) wavelet function is selected for analysis. D4 of the PRC_1 cardiac vector are regarded as reference variables $[V1^{\mathrm{PRC}_1^{\mathrm{D4}}}, V2^{\mathrm{PRC}_1^{\mathrm{D4}}}, V3^{\mathrm{PRC}_1^{\mathrm{D4}}}, V4^{\mathrm{PRC}_1^{\mathrm{D4}}}]^T$ and are used for feature derivation.

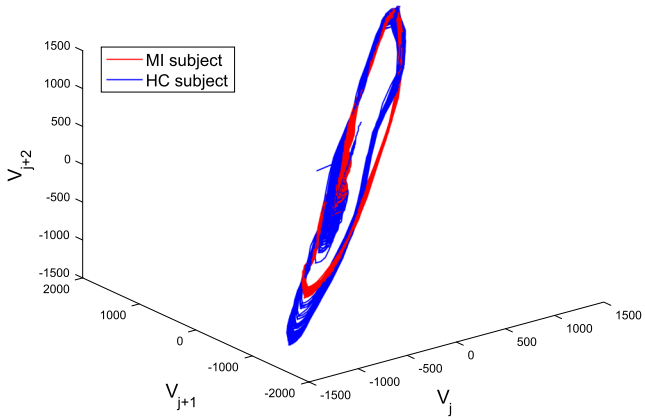
(3) Reconstruct the phase space of the reference variables with selected values of d and τ;

(4) Compute ED of 3D PSR of the reference variables. Concatenate them to form a feature vector $[ED_j^{V_1^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V_2^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V_3^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V_4^{\mathrm{PRC}_1^{\mathrm{D4}}}}]^T$.

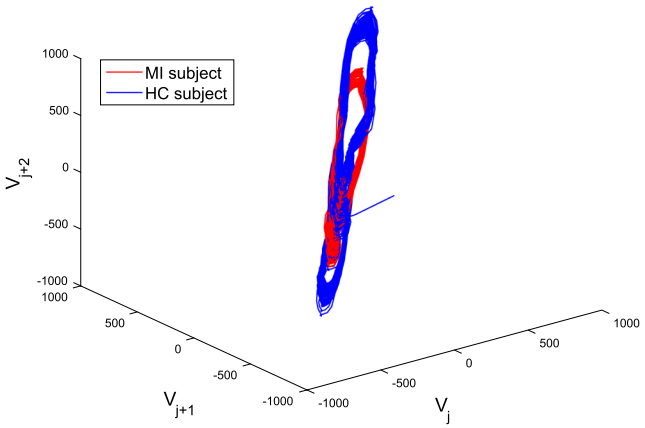
For the PTB database, cardiac vectors are analyzed and cardiac system dynamics are extracted by using ITD, DWT and 3D PSR. First, ITD of the cardiac vector is exhibited in Figs. 3 and 7. Four levels DWT of the PRC_1 of cardiac vector from the HC and MI subjects are demonstrated in 4 and 8. The db3 of the PRC_1 is utilized to form the reference variables $[V1^{\mathrm{PRC}_1^{\mathrm{D4}}}, V2^{\mathrm{PRC}_1^{\mathrm{D4}}}, V3^{\mathrm{PRC}_1^{\mathrm{D4}}}, V4^{\mathrm{PRC}_1^{\mathrm{D4}}}]^T$. Samples of the 3D PSR of the reference variables are exhibited in Fig. 5. After 3D PSR, features of $[ED_j^{V_1^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V_2^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V_3^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V_4^{\mathrm{PRC}_1^{\mathrm{D4}}}}]^T$ for cardiac vectors of the HC and MI subjects are derived through ED computation, as demonstrated in Fig. 6. As we have analyzed before, significant difference in cardiac system dynamics have been reported

Table 3 The average correlation coefficients between each PRC and original cardiac vector from HC and MI subjects in the PTB database

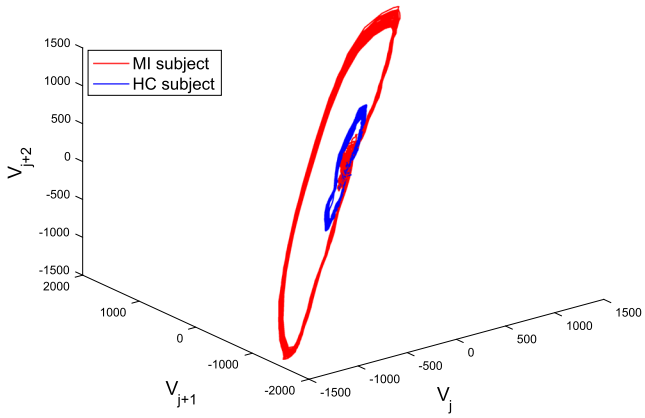
Cardiac vector type	Average correlation coefficients					
	PRC_1	PRC_2	PRC_3	PRC_4	PRC_5	PRC_6
V_1 for HC	0.6640	0.3671	0.3390	0.2873	0.1814	0.2889
V_2 for HC	0.4547	0.2704	0.2717	0.2233	0.2463	0.1556
V_3 for HC	0.7034	0.2865	0.3271	0.3165	0.1366	0.2643
V_4 for HC	0.6399	0.3025	0.2997	0.2633	0.1979	0.1732
V_1 for MI	0.5155	0.3262	0.3124	0.2474	0.2484	0.2086
V_2 for MI	0.5898	0.2809	0.2524	0.2498	0.2146	0.2073
V_3 for MI	0.6186	0.2647	0.2346	0.2153	0.2018	0.2034
V_4 for MI	0.5289	0.3190	0.2706	0.2478	0.2721	0.1972
Mean	0.5894	0.3021	0.2884	0.2563	0.2124	0.2123



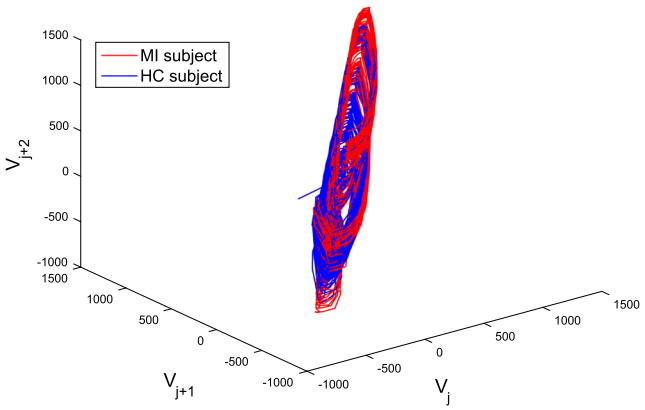
(a) 3D PSR of PRC_1^{D4} for V_1.



(b) 3D PSR of PRC_1^{D4} for V_2.



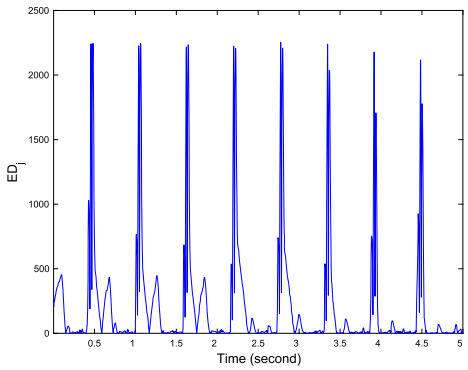
(c) 3D PSR of PRC_1^{D4} for V_3.



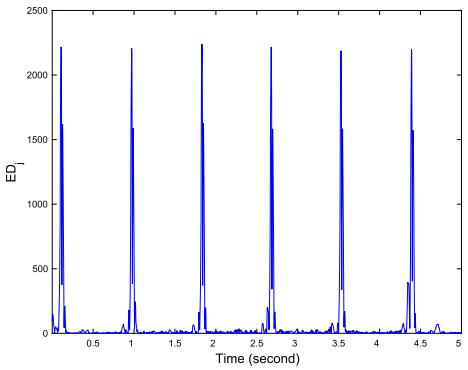
(d) 3D PSR of PRC_1^{D4} for V_4.

Fig. 5 Samples of 3D PSR of PRC_1^{D4} of 4-dimensional cardiac vector: $V = [V_1, V_2, V_3, V_4]^T$ from HC and MI subjects

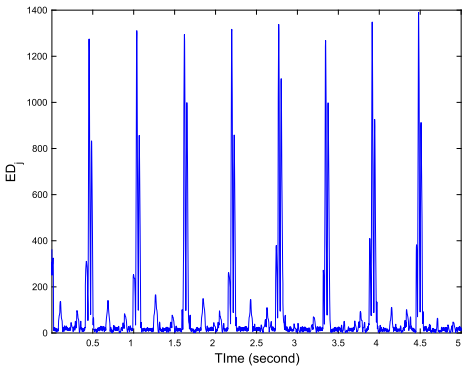
Fig. 6 Samples of the Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ of 4-dimensional cardiac vector: $V = [V_1, V_2, V_3, V_4]^T$ from HC and MI subjects



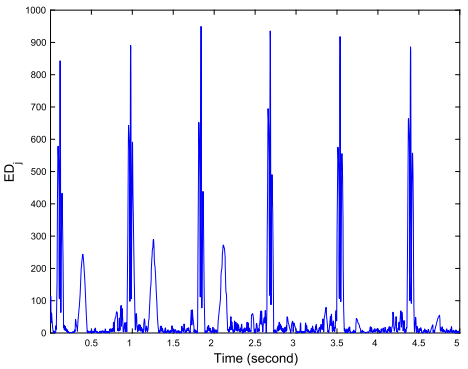
(a) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_1 from HC subject.



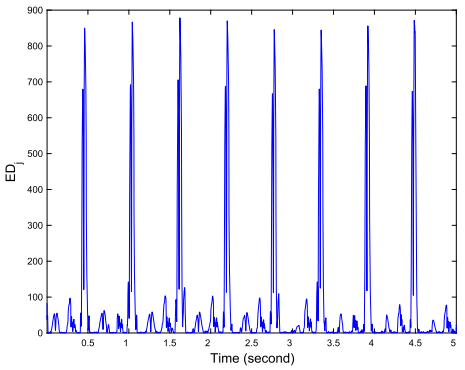
(b) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_1 from MI subject.



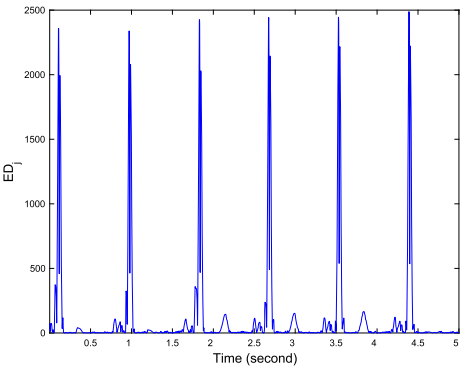
(c) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_2 from HC subject.



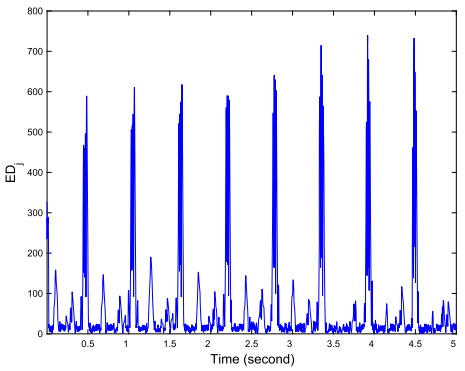
(d) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_2 from MI subject.



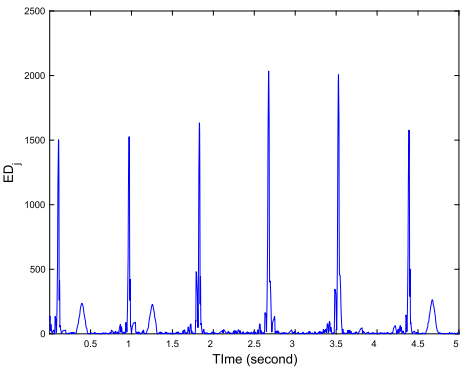
(e) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_3 from HC subject.



(f) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_3 from MI subject.



(g) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_4 from HC subject.



(h) Euclidian distance of 3D PSR of $\mathrm{PRC}_1^{\mathrm{D4}}$ for V_4 from MI subject.

Table 4 Performance of the proposed classification approach evaluated by tenfold cross-validation method

Evaluation methods	Predicted groups	Actual groups		SEN (%)	SPF (%)	ACC (%)	PPV (%)	NPV (%)	MCC
		HC	MI						
Tenfold cross-validation	HC	76	2	98.37	97.44	98.20	99.45	92.68	0.9395
	MI	6	361						

between HC and MI subjects, which can be observed obviously in Fig. 5.

Training and modeling mechanism based on selected features

In this section, we present a scheme for modeling and derivation of nonlinear cardiac system dynamics derived from the synthesis of 12-lead and Frank XYZ leads from normal and MI subjects based on the extracted features.

Consider a general nonlinear cardiac system dynamics in the following form:

$$\dot{x} = F(x;p) + v(x;p) \tag{10}$$

where $x = [x_1, \ldots, x_n]^T \in R^n$ are the system states which represent the features $[ED_j^{V_1^{PRC_1^{D4}}}, ED_j^{V_2^{PRC_1^{D4}}}, ED_j^{V_3^{PRC_1^{D4}}}, ED_j^{V_4^{PRC_1^{D4}}}]^T$, p is a constant vector of system parameters. $F(x;p) = [f_1(x;p), \ldots, f_n(x;p)]^T$ is a smooth but unknown nonlinear vector representing the cardiac system dynamics, $v(x;p)$ is the modeling uncertainty. Since the modeling uncertainty $v(x;p)$ and the cardiac system dynamics $F(x;p)$ cannot be decoupled from each other, we consider the two terms together as an undivided term, and define $\phi(x;p) := F(x;p) + v(x;p)$ as the general cardiac system dynamics. Then, the following steps are taken to model and derive the cardiac system dynamics via deterministic learning theory (Wang and Hill 2006, 2007, 2009).

In the first step, standard radial basis function (RBF) neural networks are constructed in the following form

$$f_{nn}(Z) = \sum_{i=1}^{N} w_i s_i(Z) = W^T S(Z), \tag{11}$$

where Z is the input vector, $W = [w_1, \ldots, w_N]^T \in R^N$ is the weight vector, N is the node number of the neural networks, and $S(Z) = [s_1(\| Z - \mu_1 \|), \ldots, s_N(\| Z - \mu_N \|)]^T$, with $s_i(\| Z - \mu_i \|) = \exp[\frac{-(Z-\mu_i)^T(Z-\mu_i)}{\eta_i^2}]$ being a Gaussian function, $\mu_i (i = 1, \ldots, N)$ being distinct points in state space, and η_i being the width of the receptive field.

In the second step, the following dynamical RBF neural networks are employed to model and derive the cardiac system dynamics $\phi(x;p)$:

$$\dot{\hat{x}} = -A(\hat{x} - x) + \hat{W}_j^T S_j(x) \tag{12}$$

where $\hat{x} = [\hat{x}_1, \ldots, \hat{x}_n]$ is the state vector of the dynamical RBF neural networks, $A = diag[a_1, \ldots, a_n]$ is a diagonal matrix, with $a_i > 0$ being design constants, localized RBF neural networks $\hat{W}_j^T S_j(x) = \sum_{i=1}^{N} w_{ij} s_{ij}(x)$ are used to approximate the unknown $\phi(x;p)$, where $\hat{W}_j = [w_{1j}, \ldots, w_{Nj}]^T$, $S_j = [s_{1j}, \ldots, s_{Nj}]^T$, for $j = 1, \ldots, n$.

The following law is used to update the neural weights

$$\dot{\hat{W}}_i = \dot{\tilde{W}}_i = -\Gamma_i S_i(x)\tilde{x}_i - \sigma_i \Gamma_i \hat{W}_i \tag{13}$$

where $\tilde{x}_i = \hat{x}_i - x_i$, $\tilde{W}_i = \hat{W}_i - W_i^*$, W_i^* is the ideal constant weight vector such that $\phi_i(x;p) = W_i^{*T} S_i(x) + \epsilon_i(x)$, $\epsilon_i(x) < \epsilon^*$ represents the neural network modeling error, $\Gamma_i = \Gamma_i^T > 0$, and $\sigma_i > 0$ is a small value.

With Eqs. (10)–(12), the derivative of the state estimation error $\tilde{x}_i$ satisfies

$$\dot{\tilde{x}}_i = -a_i\tilde{x}_i + \hat{W}_i^T S_i(x) - \phi_i(x;p) = -a_i\tilde{x}_i + \tilde{W}_i^T S_i(x) - \epsilon_i \tag{14}$$

In the third step, by using the local approximation property of RBF neural networks, the overall system consisting of dynamical model (14) and the neural weight updating law (13) can be summarized into the following form in the region Ω_ζ

$$\begin{bmatrix} \dot{\tilde{x}}_i \\ \dot{\tilde{W}}_{\zeta i} \end{bmatrix} = \begin{bmatrix} -a_i & S_{\zeta i}(x)^T \\ -\Gamma_{\zeta i} S_{\zeta i}(x) & 0 \end{bmatrix} \begin{bmatrix} \tilde{x}_i \\ \tilde{W}_{\zeta i} \end{bmatrix} + \begin{bmatrix} -\epsilon_{\zeta i} \\ -\sigma_i \Gamma_{\zeta i} \hat{W}_{\zeta i} \end{bmatrix} \tag{15}$$

and

$$\dot{\hat{W}}_{\bar{\zeta} i} = \dot{\tilde{W}}_{\bar{\zeta} i} = -\Gamma_{\bar{\zeta} i} S_{\bar{\zeta}}(x)\tilde{x}_i - \sigma_i \Gamma_{\bar{\zeta} i} \hat{W}_{\bar{\zeta} i} \tag{16}$$

where $\epsilon_{\zeta i} = \epsilon_i - \tilde{W}_{\bar{\zeta} i}^T S_{\bar{\zeta}}(x)$. The subscripts $(\cdot)_\zeta$ and $(\cdot)_{\bar{\zeta}}$ are used to stand for terms related to the regions close to and far away from the trajectory $\varphi_\zeta(x_0)$. The region close to the trajectory is defined as $\Omega_\zeta := \{Z | \text{dist}(Z, \varphi_\zeta) \le d_\iota\}$, where $Z = x, d_\iota > 0$ is a constant satisfying $s(d_\iota) > \iota$, $s(\cdot)$ is the RBF used in the network, ι is a small positive constant. The related subvectors are given as: $S_{\zeta i}(x) = [s_{j1}(x), \ldots, s_{j\zeta}(x)]^T \in R^{N_\zeta}$, with the neurons centered in the local region Ω_ζ, and $W_\zeta^* = [w_{j1}^*, \ldots, w_{j\zeta}^*]^T \in R^{N_\zeta}$ is the

corresponding weight subvector, with $N_\zeta < N$. For localized RBF neural networks, $|\tilde{W}_{\bar{\zeta}i}^T S_{\bar{\zeta}i}(x)|$ is small, so $\epsilon_{\zeta i} = O(\epsilon_i)$.

By the convergence result, we can obtain a constant vector of neural weights according to

$$\bar{W}_i = mean_{t\in[t_a,t_b]}\hat{W}_i(t) \tag{17}$$

where $t_b > t_a > 0$ represent a time segment after the transient process. Therefore, we conclude that accurate identification of the function $\phi_i(x;p)$ is obtained along the trajectory $\varphi_\zeta(x_0)$ by using $\bar{W}_i^T S_i(x)$, i.e.,

$$\phi_i(x;p) = \bar{W}_i^T S_i(x) + \epsilon_{i2} \tag{18}$$

where $\epsilon_{i2} = O(\epsilon_{i1})$ and subsequently $\epsilon_{i2} = O(\epsilon^*)$.

Classification mechanism

In this section, we present a scheme to classify normal and MI cardiac vectors.

Consider a training dataset consisting of cardiac system patterns φ_ζ^k, $k = 1,\ldots,M$, with the *kth* training pattern φ_ζ^k generated from

$$\dot{x} = F^k(x;p^k) + v^k(x;p^k), \quad x(t_0) = x_{\zeta 0} \tag{19}$$

where $F^k(x;p^k)$ denotes the cardiac system dynamics, $v^k(x;p^k)$ denotes the modeling uncertainty, p^k is the system parameter vector.

As demonstrated in Sect. 2.7, the general cardiac system dynamics $\phi^k(x;p^k) := F^k(x;p^k) + v^k(x;p^k)$ can be accurately derived and preserved in constant RBF neural networks $\bar{W}^{k^T} S(x)$. By utilizing the learned knowledge obtained in the training stage, a bank of M estimators is constructed for the training cardiac system patterns as follows:

$$\dot{\bar{\chi}}^k = -B(\bar{\chi}^k - x) + \bar{W}^{k^T} S(x) \tag{20}$$

where $k = 1,\ldots,M$ is used to stand for the *kth* estimator, $\bar{\chi}^k = [\bar{\chi}_1^k,\ldots,\bar{\chi}_n^k]^T$ is the state of the estimator, $B = diag[b_1,\ldots,b_n]$ is a diagonal matrix which is kept the same for all estimators, x is the state of an input test cardiac system pattern generated from Eq. (10).

In the classification phase, by comparing the test cardiac system pattern (standing for a HC or MI cardiac system pattern) generated from the cardiac system (10) with the set of M estimators (20), we obtain the following test error systems:

$$\begin{aligned} &\dot{\tilde{\chi}}_i^k = -b_i\tilde{\chi}_i^k + \bar{W}_i^{k^T} S_i(x) - \phi_i(x;p), \\ &i = 1,\ldots,n, \quad k = 1,\ldots,M \end{aligned} \tag{21}$$

where $\tilde{\chi}_i^k = \bar{\chi}_i^k - x_i$ is the state estimation (or synchronization) error. We compute the average L_1 norm of the error $\tilde{\chi}_i^k(t)$

$$\|\tilde{\chi}_i^k(t)\|_1 = \frac{1}{\mathrm{T}_c}\int_{t-\mathrm{T}_c}^{t} |\tilde{\chi}_i^k(\tau)|d\tau, \quad t \geq \mathrm{T}_c \tag{22}$$

where T_c is the cycle of cardiac vector.

The fundamental idea of the classification between HC and MI cardiac vectors is that if a test cardiac system pattern is similar to the trained cardiac system pattern s ($s \in \{1,\ldots,k\}$), the constant RBF network $\bar{W}_i^{s^T} S_i(x)$ embedded in the matched estimator s will quickly recall the learned knowledge by providing accurate approximation to the cardiac system dynamics. Thus, the corresponding error $\|\tilde{\chi}_i^s(t)\|_1$ will become the smallest among all the errors $\|\tilde{\chi}_i^k(t)\|_1$. Based on the smallest error principle, the appearing test cardiac system pattern can be classified. We have the following classification scheme.

Classification scheme If there exists some finite time t^s, $s \in \{1,\ldots,k\}$ and some $i \in \{1,\ldots,n\}$ such that $\|\tilde{\chi}_i^s(t)\|_1 < \|\tilde{\chi}_i^k(t)\|_1$ for all $t > t^s$, then the appearing cardiac system pattern can be classified and MI can be detected.

The classification performance will be evaluated with the tenfold cross-validation style in which the variance of the estimate for the classifiers is reduced. The data are divided into the training and test subsets. For the tenfold cross-validation, the data set is divided into ten subsets. Each time, one of the ten subsets is used as the test set and the other nine subsets are put together to form a training set. As such, every fold has been used nine times as train data and one time as test data. The final result is the average of the 10 implementations. For the evaluation, six performance parameters are used including the Sensitivity (SEN), the Specificity (SPF), the Accuracy (ACC), the Positive Predictive Value (PPV), the Negative Predictive Value (NPV) and the Matthews Correlation Coefficient (MCC). These measurements are defined as follows (Azar and El-Said 2014):

$$\mathrm{SEN} = \frac{TP}{TP+FN} \times 100(\%) \tag{23}$$

$$\mathrm{SPF} = \frac{TN}{TN+FP} \times 100(\%) \tag{24}$$

$$\mathrm{ACC} = \frac{TP+TN}{TP+TN+FN+FP} \times 100(\%) \tag{25}$$

$$\mathrm{PPV} = \frac{TP}{TP+FP} \times 100(\%) \tag{26}$$

$$\mathrm{NPV} = \frac{TN}{TN+FN} \times 100(\%) \tag{27}$$

$$\mathrm{MCC} = \frac{TP\times TN - FN\times FP}{\sqrt{(TP+FN)(TP+FP)(TN+FN)(TN+FP)}} \tag{28}$$

Table 5 Summary of classification performance on the HC and MI subjects obtained from the same PTB database in the literature

References	Number of leads	Feature extraction	Classifiers	Tenfold cross-validation	Performance
Sun et al. (2012)	12 leads	ST segments and R-R intervals	LTMIL	Yes	Sensitivity = 92.6%
Dohare et al. (2018)	12 leads	ECG wave components and clinical wave intervals	SVM	Yes	Accuracy = 98.33%
Banerjee and Mitra (2013)	One lead (lead III)	Using XWT to yield WCS and WCOH	Threshold based classifier	No	Accuracy = 97.6%
Arif et al. (2010)	12 leads	Using PCA to extract 117 features from ST-T segment and Q wave	back propagation neural network	No	Accuracy = 93.7%
Al-Kindi et al. (2011)	12 leads	R complexes time index extracted from Q, R, S waves and J points	Not mentioned	No	Sensitivity = 85%
Liu et al. (2015)	12 leads	20th order polynomial function measured by the Akaike information criterion	J48 decision tree	No	Accuracy = 93.7%
Sharma et al. (2015)	12 leads	Multiscale wavelet energies and eigenvalues of multiscale covariance matrices	SVM with RBF kernel	No	Accuracy = 96%
Sharma and Sunkaria (2018)	6 leads	Sample entropy, normalized sub-band energy, log energy entropy and median slope	SVM	Yes	Accuracy = 98.84%
Acharya et al. (2017e)	One lead (lead II)	End-to-end learning structure without any hand-crafted feature extraction	CNN	Yes	Accuracy = 95.22%
Strodthoff and Strodthoff (2019)	15 leads	End-to-end learning structure without any hand-crafted feature extraction	CNN	Yes	Accuracy = 96.1%
Lih et al. (2020)	One lead (lead II)	Automatic feature extraction and selection with deep learning method	CNN-LSTM	Yes	Accuracy = 98%
Proposed work	15 leads	Extracted through ITD, DWT, PSR and ED	Dynamical estimators	Yes	Accuracy = 98.20%

where TP is the number of true positives, FN is the number of false negatives, TN is the number of true negatives and FP is the number of false positives. To be accurate, a classifier must have a high classification accuracy, a high sensitivity, as well as a high specificity (Chu 1999). For a larger value of MCC, the classifier performance will be better (Azar and El-Said 2014; Yuan et al. 2007).

Results

Experiments are implemented using matlab software and tested on an Intel Core i7 6700K 3.5GHz computer with 64GB RAM. We assign feature vector sequences for all the heart beat records of HC and MI subjects in the PTB database. According to the method described in Sect. 2.6, we extract features, which means the input of the RBF neural networks $[ED_j^{V1^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V2^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V3^{\mathrm{PRC}_1^{\mathrm{D4}}}}, ED_j^{V4^{\mathrm{PRC}_1^{\mathrm{D4}}}}]^T$. In order to eliminate data difference between different features, all feature data are normalized to $[-1, 1]$. Several experiments are carried out to verify the effectiveness of the proposed method on the 78 HC and 367 MI heart beat records. The classification results on HC an MI subjects have been illustrated in Table 4 with tenfold cross-validation style. Our study demonstrates the accuracy improvements to differentiate between HC and MI. Overall, our classification approach achieves good performance, which indicates that the proposed pattern recognition system can effectively classify MI by using nonlinear features and neural network based classification tools.

Discussion

Experimental results of this study revealed that MI subjects could be differentiated from HC subjects automatically by means of nonlinear features and neural networks based upon hybrid signal processing methods. The proposed scheme focuses not only on providing evidence to support the claim that MI subjects demonstrate altered cardiac system dynamics in comparison to HC subjects, but also on providing an automatic and objective method to detect MI cardiac vectors.

Sun et al. (2012) proposed a latent topic multiple instance learning (LTMIL) algorithm for MI detection based on time-domain features of ST segments and R-R intervals from ECG. Their method obtained a sensitivity of 92.6% with 74 features. Dohare et al. (2018) used four time-domain features including P duration, QRS duration, ST-T complex interval and QT interval, which were globally determined from average beats of all the 12-lead ECG signals. Then peak to peak amplitude, area, mean,

standard deviation, skewness and kurtosis were computed for these features. To reduce the computational complexity, principal component analysis (PCA) reduction technique was adopted for the feature reduction. The accuracy with SVM classifier for the MI detection was reported to be 98.33%. Banerjee and Mitra (2013) employed cross wavelet transform (XWT) to produce wavelet cross spectrum (WCS) and wavelet coherence (WCOH) for the analysis of spectral differences between HC and MI subjects. The reported accuracy with threshold based classifier was 97.6% based on lead-III ECG signals. Arif et al. (2010) extracted 117-dimensional features from ST-T region and Q wave region of each beat by using lead-wise principal components analysis (PCA) . Features were derived from 12-lead ECG signals and fed into the back propagation neural networks (BPNN) classifier for classification, which was with the reported accuracy of 93.7%. Al-Kindi et al. (2011) used R-complex time index detection method to extract features from Q, R, S waves and J points, which could serve as an clinical indicator of MI. 12-lead ECG signals were derived from 20 normal and 20 MI subjects and the reported sensitivity was 85% for the detection of MI subjects. Liu et al. (2015) fitted a given ECG signal with a 20th order polynomial function to derive new ECG features. They reported 94.4% classification accuracy with J48 decision tree model for the diagnosis of MI. Sharma et al. (2015) calculated multiscale wavelet energies and eigenvalues of multiscale covariance matrices to act as as diagnostic features. The reported accuracy with the SVM RBF kernel classifier was 96%. Sharma and Sunkaria (2018) calculated sample entropy, normalized sub-band energy, log energy entropy and median slope over selected bands of 6-lead ECG signals. These vectors were used as features and fed to the SVM classifier for the classification of MI and HC ECG signals. The reported accuracy was 98.84%. Acharya et al. (2017e) constructed an end-to-end learning structure without any hand-crafted feature extraction by using a single lead ECG signal. The reported accuracy for the automated detection of normal and MI ECG beats (with noise and without noise) by using deep learning algorithm (CNN) was 93.53% and 95.22%, respectively. Strodthoff and Strodthoff (2019) also built an end-to-end learning structure without any hand-crafted feature extraction by including different combinations of leads, which could estimate the relative amount of information that these channels contribute to the classification decision. The reported accuracy for the automated detection of normal and MI ECG beats by using CNN was 96.1%. Lih et al. (2020) proposed combined 16-layer CNN and Long Short-Term Memory (LSTM) models for the detection of MI with automatic feature extraction and selection. The reported accuracy was 98%.

Different from the above discussed methods, this study proposes a hybrid method to extract nonlinear features using ITD, DWT, PSR and ED tools based on the synthesis of traditional 12-lead and Frank XYZ leads. These features are fed into dynamical estimators consisting of constant RBF neural networks to classify MI and HC subjects. Comparison of the classification performance to other state-of-the-art methods on the same database is demonstrated in Table 5. The proposed method provides an average classification accuracy of 98.20% through tenfold cross-validation style. Due to the use of tenfold cross-validation, the classification performance is robust. The method studied in this paper has the potential to serve as a supportive technical means to other approaches such as coronary arteriography and cardiac ultrasound for the diagnosis of MI. Although our classification accuracy is not higher than that reported in (Sharma and Sunkaria 2018), we present a new classification tool together with building a novel feature vector rather than using directly the ECG signals. In addition, we used 4 features which are less than the 10 features used in (Sharma and Sunkaria 2018). Modeling, identification and classification of cardiac system dynamics were employed instead of putting feature vectors directly into the classifier in comparison to other methods. This provides another candidate tool for the detection of MI. In the present study, we apply the extracted features to the SVM classifier with RBF Kernel and CNN classifier and achieve the classification accuracy of 89.66% and 94.38%, respectively, by using tenfold cross-validation. It seems that the accuracy is lower than the result reported in (Sharma and Sunkaria 2018) and (Acharya et al. 2017e) with the same kinds of classifiers. Owing to the working principle of the learning and classification algorithm mentioned in the present study, our proposed features may not work well when fitting with other classifiers.

Parameters in physiological models are often uncertain due to either measurement uncertainty and/or natural physiological variability (Pathmanathan et al. 2019). Comparison methods with deep learning usually used hyperparameters to study increasing levels of parametric uncertainty. They studied model robustness by increasing parameter uncertainty, and explored when and how new model behaviors (such as repolarization failure or loss of spike-and-dome morphology) occured. Strodthoff and Strodthoff (2019) reported ensemble scores corresponding to models with different random initializations without performing any form of hyperparameter tuning or model selection using test data. For calculating the ensemble score, they combined five identical models and reported the ensemble score formed by averaging the predicted scores after the softmax layer. Their classifier reached 93.3% sensitivity, 89.7% specificity and 96.1% accuracy. Sharma and Sunkaria (2018) calculated sample entropy, which is a measure of the level of uncertainty or randomness of a signal. The entropy can present the uncertainty and irregularity of corresponding signals. Furthermore, it reflects

quantitatively the overall distribution trend of signal by statistical analysis towards the information hidden in the signals (Han and Shi 2019). Entropy calculated at different wavelet bands can provide useful information about the pathology of the underlying dynamic system. The human heart is a complex dynamic system, and hence entropy can be used to analyze the associated cardiac complexities. They achieved the sensitivity, specificity and accuracy of 99.35%, 98.29% and 98.84%, respectively. Other comparison methods in Table 5 did not consider or report the effect of parametric uncertainty on the classification performance. In the present study, assessments of the relationship between the level of the PRCs, embedding dimension, time lag, parameters in ITD and DWT, and the classification accuracy can also be considered in future investigations. It would be of interest to develop strategy for adaptive selection of PSR parameters and figure out the optimal solutions, which could create best classification performance. These are topics for future work.

There are many factors in the framework of the proposed method that facilitate the satisfactory classification performance, which include the following key points. (1) The ability to reveal the hidden characteristics present in the ECG signals for the MI detection. Synthesis of 12-lead and Frank XYZ leads to build a new 4-dimensional cardiac vector may fully reflect the pathological alterations in the abnormal patterns of cardiac system dynamics provoked by MI. ITD could extract most important information of the cardiac vector through predominant PRCs. DWT decomposes the predominant PRCs into different frequency bands, which are used to construct the reference variables. (2) Clear depiction of the behaviour of the cardiac system. PSR plots cardiac system dynamics along the advisable db3 sub-bands (D4) of predominant PRCs trajectory in a 3D phase space diagram and visualizes the cardiac system dynamics, which is demonstrated in Fig. 5. This implies that the full dynamics of the cardiac system are accessible in this space, and for this reason, features extracted from it can potentially contain more and/or different information than the common features extraction method. ED measures and derives features. (3) Less computational complexity. The proposed approach in this study is a completely end-to-end structure without ST segment evaluation, which is different from traditional approaches. (4) Neural networks that cooperate with deterministic learning could model and identify nonlinear cardiac system dynamics between HC and MI subjects. MI detection is based on the disparity of cardiac system dynamics between the two groups.

Analysis of ST-segment elevation, as well as the high apex and inversion of T waves, and the appearance of pathological Q waves, is substantially more popular than the analysis of long-duration ECG signal fragments, which is error-prone and may not achieve satisfactory diagnostic performance due to high beat-to-beat variability of these features among individuals. In this study, we propose a completely end-to-end structure to analyze long-interval ECG signal without ST segment evaluation. ITD, DWT, PSR and ED are combined to extract features from the synthesis of 12-lead and Frank XYZ leads of ECG signals. Features are mapped to neural networks to model, identify and classification of cardiac system dynamics. Considering the complicated relationship between the distinguished patterns of healthy and MI ECG signals, it is suggested that for clinical practice, in addition to the ST segment evaluation, discrimination of the disparity of cardiac system dynamics between healthy and MI ECG signals should be another candidate for MI detection. This is expected to result in the selection of nonlinear and dynamic ECG features, which may reveal the working principle of the abnormal patterns in pathological ECG signals and improve the MI detection outcome.

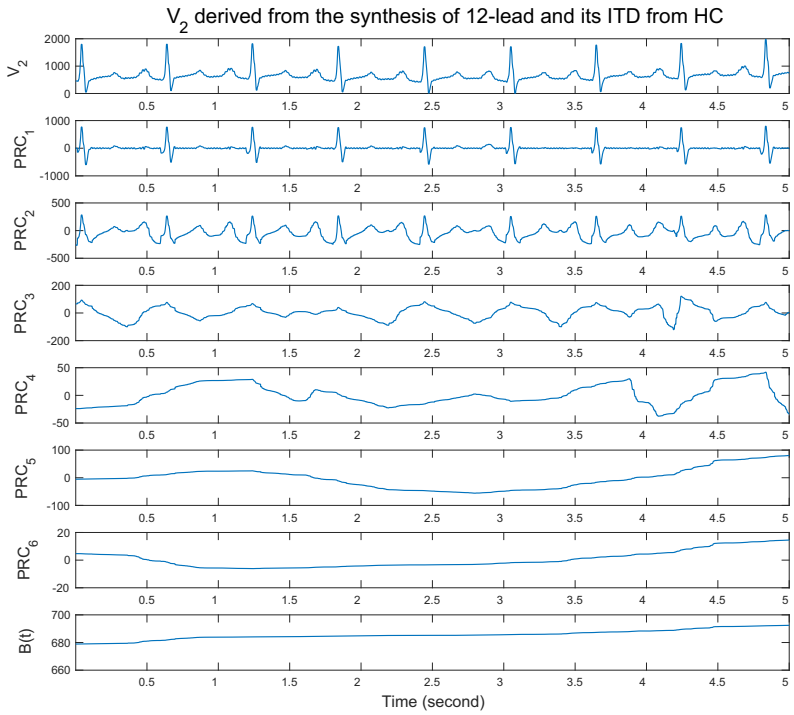
Conclusions

In this study, a novel feature extraction technique including ITD, DWT, PSR and ED is proposed for detection of myocardial infarction, which is computationally simple and easy to implement. The results of this study indicate that the pattern classification of cardiac vector based on the synthesis of traditional 12-lead and Frank XYZ leads can offer an objective method to assess the disparity of cardiac system dynamics between HC and MI subjects. However, some limitations such as the relatively small size of the database, the regulation principle of the embedding dimension and time lag, still need to be improved and overcome. Future work will include a clinical validation of the proposed technique with a larger number of patients. Features introduced in other methods such as various entropies, Hurst exponent, fractal dimension and other nonlinear features, can also be explored in the proposed framework to evaluate its classification performance. The proposed automated MI detection system can assist cardiologists in cross-checking their diagnosis. In addition, it can be extended to grade the severity of MI.

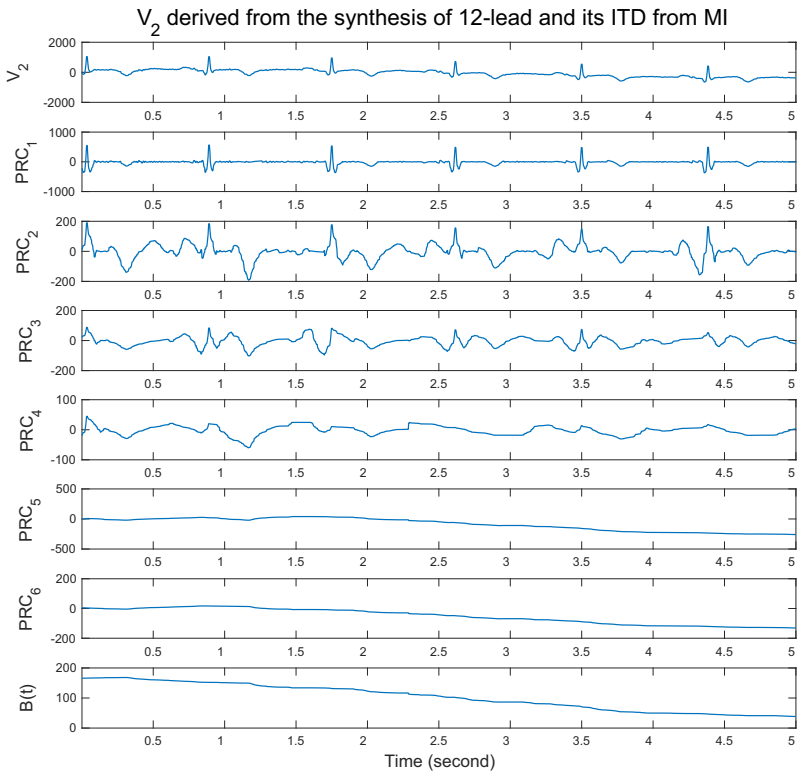
Appendix

See Figs. 7 and 8.

Fig. 7 Samples of the ITD of the subvectors V_2, V_3 and V_4 from HC and MI subject

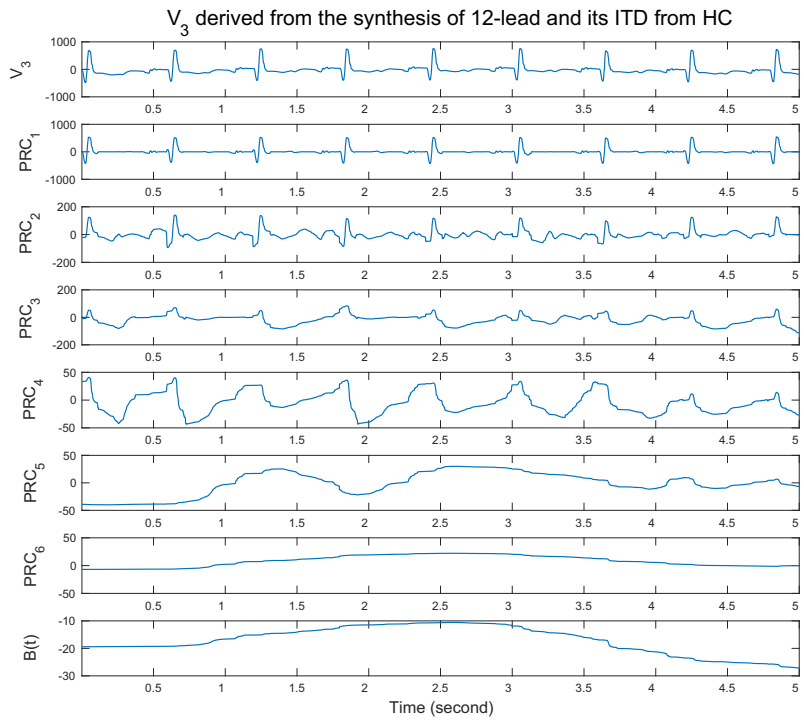


(a) Original V_2 vector and its ITD from HC.

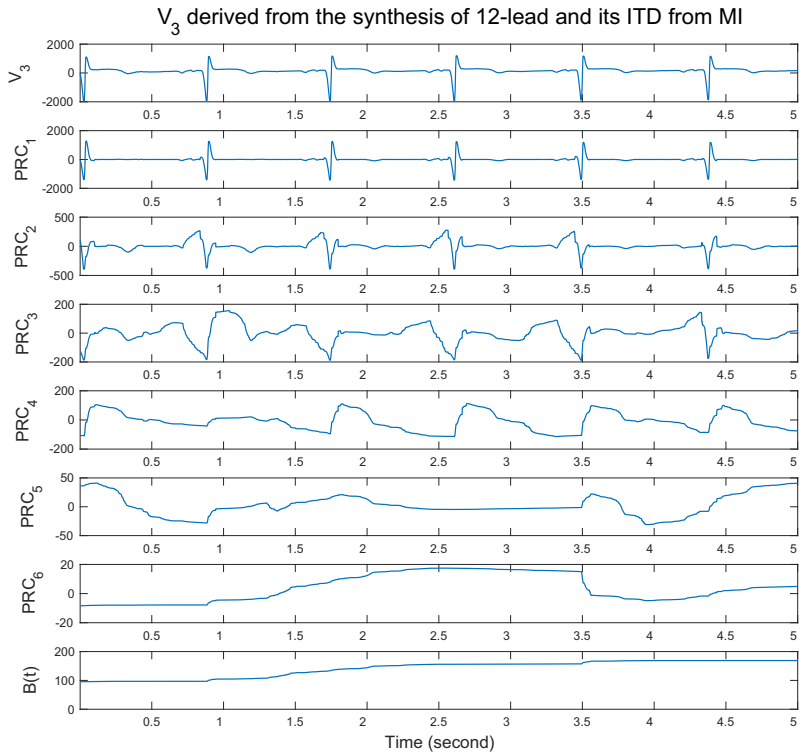


(b) Original V_2 vector and its ITD from MI.

Fig. 7 continued

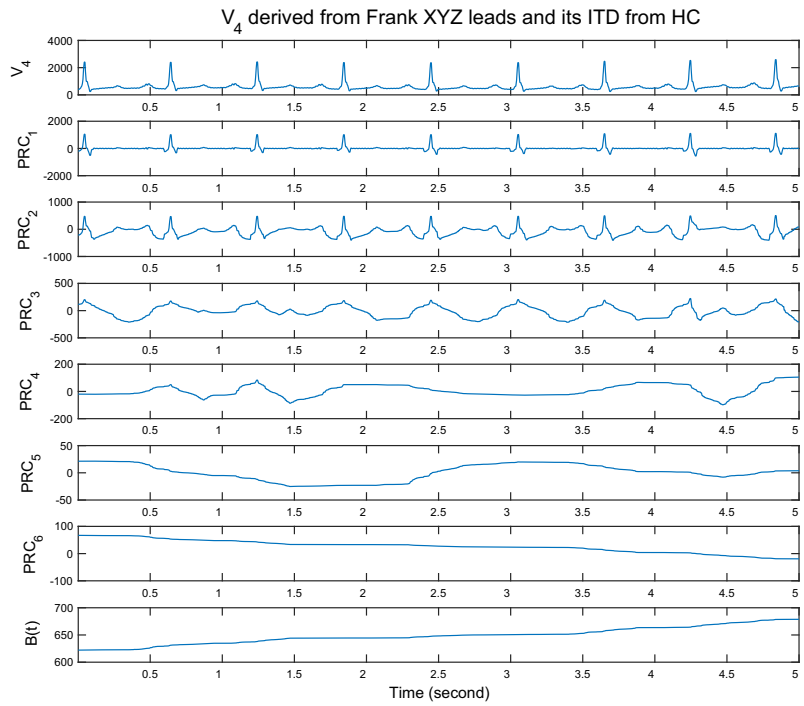


(c) Original V_3 vector and its ITD from HC.

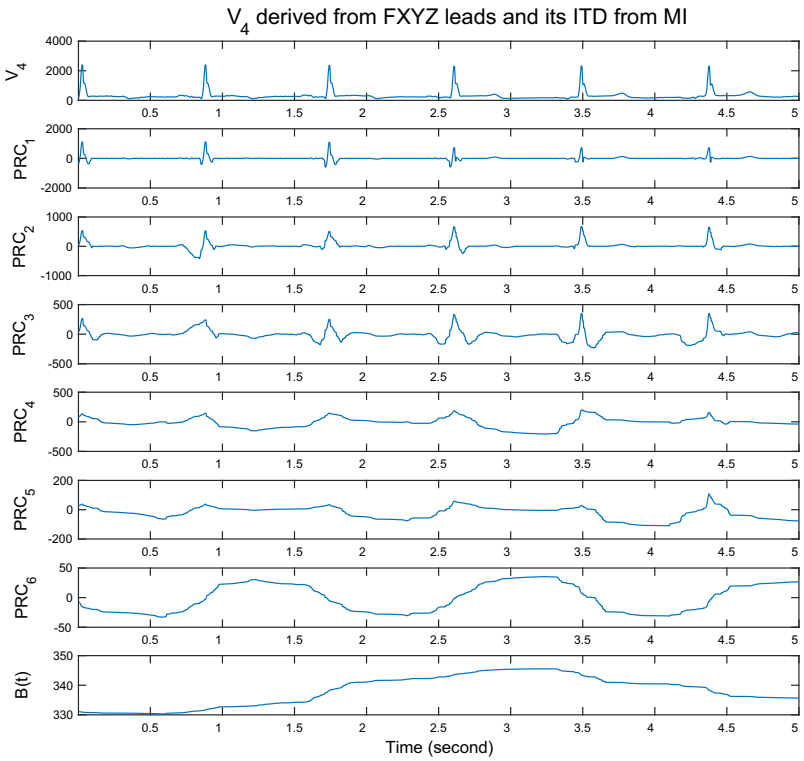


(d) Original V_3 vector and its ITD from MI.

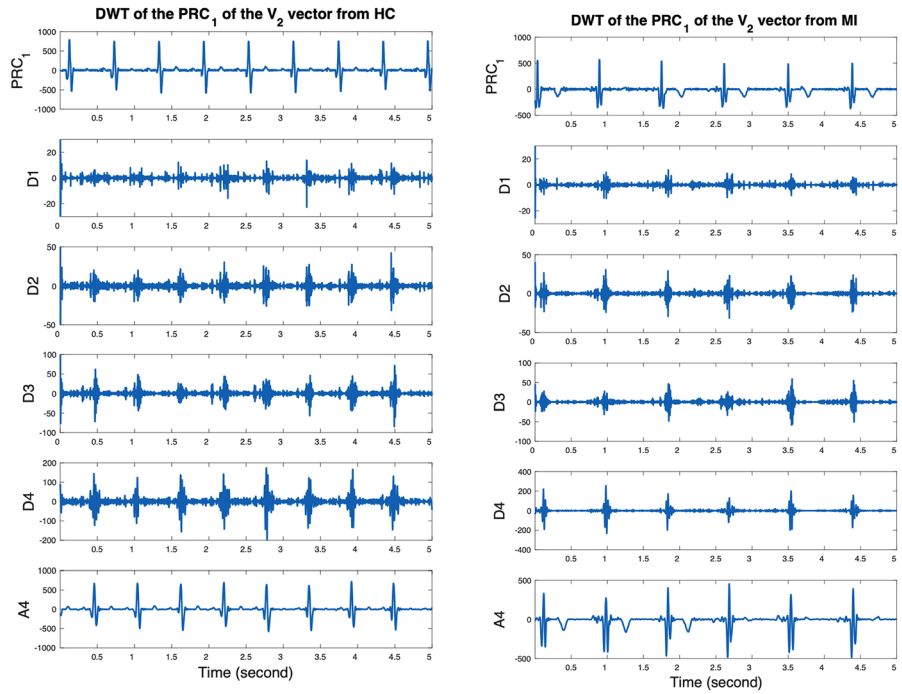
Fig. 7 continued



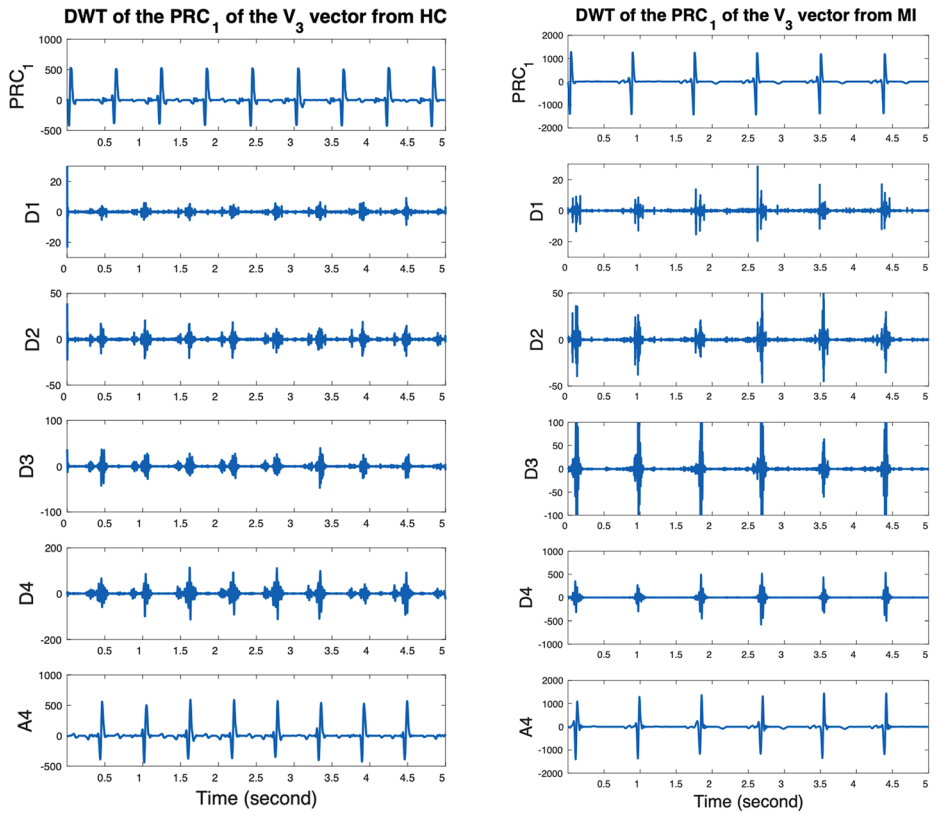
(e) Original V_4 vector and its ITD from HC.



(f) Original V_4 vector and its ITD from MI.

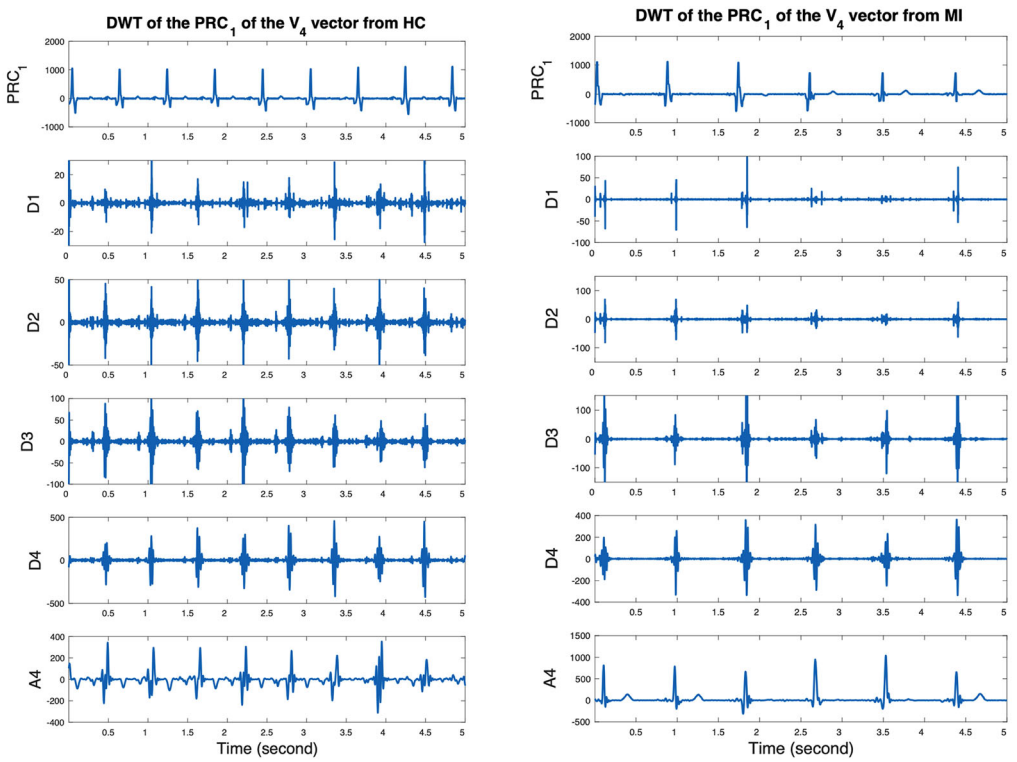


(a) DWT of PRC_1 of V_2 from HC subject. **(b)** DWT of PRC_1 of V_2 from MI subject.



(c) DWT of PRC_1 of V_3 from HC subject. **(d)** DWT of PRC_1 of V_3 from MI subject.

Fig. 8 Samples of four levels DWT of PRC_1 of the subvector V_2, V_3 and V_4 from HC and MI subjects



(e) DWT of PRC_1 of V_4 from HC subject. **(f)** DWT of PRC_1 of V_4 from MI subject.

Fig. 8 continued

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Data availibility statement The datasets used and/or analyzed during in current study are available from the public Physionet PTB dataset (Bousseljot et al. 1995; Goldberger et al. 2000) which can be accessed at the website: https://www.physionet.org/content/ptbdb/1.0.0/.

Declarations

Conflict of interest There is no conflict of interest.

Ethical approval There is no issue with Ethical approval and Informed consent.

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